WASP-1b

R-1449

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-1_131220` · created 2026-08-01T09:21:48+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456646.666 +/- 0.005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.097 +/- 0.028
Transit depth (Rp/Rs)^2	0.94 +/- 0.54 [%]
Orbital Inclination (inc)	85.13 +/- 1.47
Airmass coefficient 1 (a1)	0.959 +/- 0.013
Airmass coefficient 2 (a2)	0.0296 +/- 0.0092
Scatter in the residuals of the lightcurve fit is	1.38 %
Best Comparison Star	#1 - [293, 377]
Optimal Aperture	3.35
Optimal Annulus	7.35
Transit Duration (day)	0.137 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.097 ± 0.028 vs 0.1036 (deviation 0.24σ)
Duration fit vs published	0.137 d vs 0.1567 d (deviation 1.64σ)
Mid-time O–C	13.3 min
Depth SNR	3.5
Residual scatter	1.38 %

Light curve

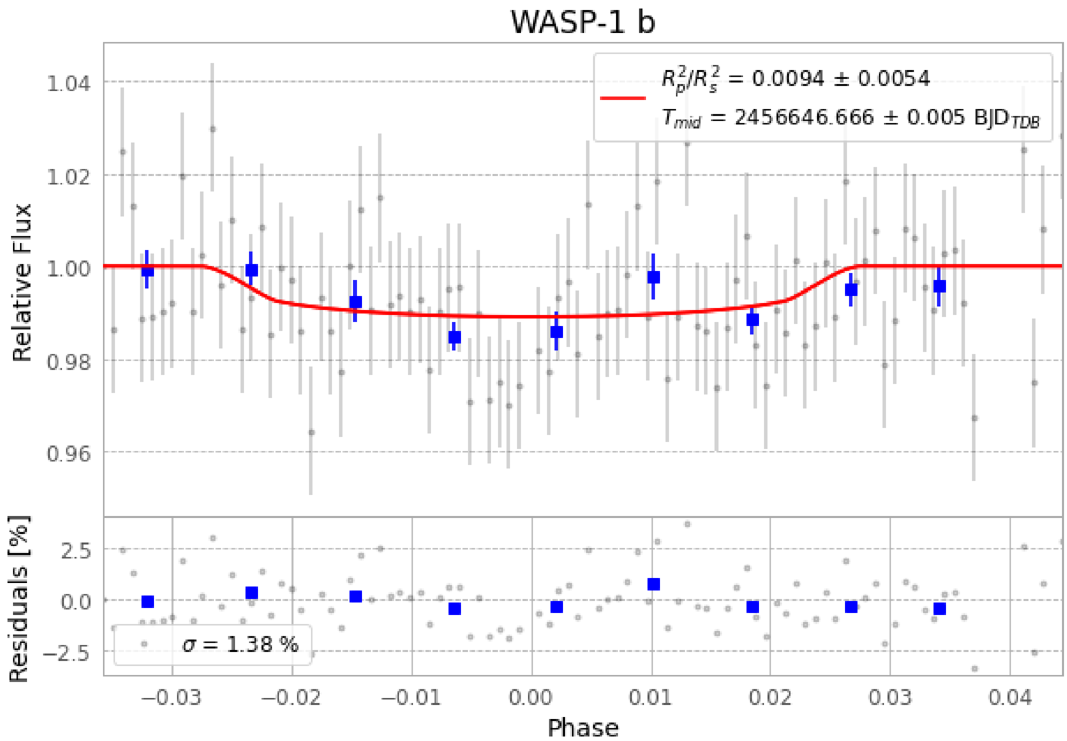

Figure 1 — FinalLightCurve_WASP-1 b_2013-12-19.png (SHA-256 7ea26eceb8454615…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [405, 322], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 e903fe8e096a596d…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

98 FITS frames (64 MB), `WASP-1131220014512.FITS` → `WASP-1131220063612.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-131220.log (2c190a6d7efd…), FinalLightCurve_WASP-1 b_2013-12-19.png (7ea26eceb845…), FinalLightCurve_WASP-1 b_2013-12-19.csv (afcbeb49e47c…)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

